

# Multi-Source Candidate Data Transformer – Technical Design

The system deterministically transforms candidate data from ATS JSON and Resume PDF into a single canonical profile. It emphasizes explainability through provenance, fixed-rule confidence scoring, deterministic conflict resolution, and runtime-configurable output projection.

## Pipeline

ATS JSON + Resume PDF → Parse & Validate → Extract Candidate Facts → Normalize Values → Merge & Conflict Resolution → Provenance & Confidence → Runtime Projection → Output Validation → Final JSON

## Canonical Schema & Normalization

| Field            | Normalization                              |
|------------------|--------------------------------------------|
| Phones           | E.164                                      |
| Experience Dates | YYYY-MM                                    |
| Country          | ISO-3166 alpha-2                           |
| Skills           | Canonical skill names                      |
| Strings          | Trim, whitespace collapse, stable ordering |

## Merge & Confidence Strategy

Facts are normalized and grouped by canonical field. Conflicts are resolved deterministically using: (1) source precedence (ATS > Resume), (2) higher field confidence, (3) extraction quality, and (4) lexical tie-break. Losing evidence is retained in provenance. Field confidence = Source Reliability × Extraction Quality × Agreement Factor. Overall confidence is a weighted aggregation of key fields.

## Runtime Projection

The canonical profile is the single source of truth. A runtime configuration selects fields, remaps paths, applies optional normalization, includes/excludes provenance and confidence, and enforces missing-value policies (null, omit, or error). The projected output is validated before emission.

## Edge Cases & Out of Scope

Edge cases: malformed ATS JSON, corrupt Resume PDF, conflicting phones, skill synonyms, and missing required projected fields. Out of scope: NLP/LLM inference, additional source types, learned confidence models, distributed processing, and rich UI.